MSG No. 70.530 $Fd'd'd$ [Type III, orthorhombic]

Table 1: Wyckoff site: `8a`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{8}\right]$	`[1,13,22,26]`
2	$\left[\frac{5}{8}, \frac{5}{8}, \frac{1}{8}\right]$	`[2,14,21,25]`
3	$\left[\frac{7}{8}, \frac{7}{8}, \frac{7}{8}\right]$	`[3,15,24,28]`
4	$\left[\frac{3}{8}, \frac{3}{8}, \frac{7}{8}\right]$	`[4,16,23,27]`
5	$\left[\frac{1}{8}, \frac{5}{8}, \frac{5}{8}\right]$	`[5,9,18,30]`
6	$\left[\frac{5}{8}, \frac{1}{8}, \frac{5}{8}\right]$	`[6,10,17,29]`
7	$\left[\frac{7}{8}, \frac{3}{8}, \frac{3}{8}\right]$	`[7,11,20,32]`
8	$\left[\frac{3}{8}, \frac{7}{8}, \frac{3}{8}\right]$	`[8,12,19,31]`

Table 2: Wyckoff site: `8b`, site symmetry: `2'2'2`

No.	position	mapping
1	$\left[\frac{1}{8}, \frac{1}{8}, \frac{5}{8}\right]$	`[1,13,22,26]`
2	$\left[\frac{5}{8}, \frac{5}{8}, \frac{5}{8}\right]$	`[2,14,21,25]`
3	$\left[\frac{7}{8}, \frac{7}{8}, \frac{3}{8}\right]$	`[3,15,24,28]`
4	$\left[\frac{3}{8}, \frac{3}{8}, \frac{3}{8}\right]$	`[4,16,23,27]`
5	$\left[\frac{1}{8}, \frac{5}{8}, \frac{1}{8}\right]$	`[5,9,18,30]`
6	$\left[\frac{5}{8}, \frac{1}{8}, \frac{1}{8}\right]$	`[6,10,17,29]`
7	$\left[\frac{7}{8}, \frac{3}{8}, \frac{7}{8}\right]$	`[7,11,20,32]`
8	$\left[\frac{3}{8}, \frac{7}{8}, \frac{7}{8}\right]$	`[8,12,19,31]`

Table 3: Wyckoff site: `16c`, site symmetry: `-1`

No.	position	mapping
1	$[0, 0, 0]$	`[1,3]`
2	$\left[\frac{3}{4}, \frac{3}{4}, 0\right]$	`[2,28]`
3	$\left[\frac{1}{4}, \frac{1}{4}, 0\right]$	`[4,26]`
4	$\left[0, \frac{3}{4}, \frac{3}{4}\right]$	`[5,15]`
5	$\left[\frac{3}{4}, 0, \frac{3}{4}\right]$	`[6,24]`
6	$\left[0, \frac{1}{4}, \frac{1}{4}\right]$	`[7,13]`
7	$\left[\frac{1}{4}, 0, \frac{1}{4}\right]$	`[8,22]`
8	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[9,11]`
9	$\left[\frac{3}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[10,20]`
10	$\left[\frac{1}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[12,18]`
11	$\left[\frac{3}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[14,32]`
12	$\left[\frac{1}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[16,30]`
13	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[17,19]`
14	$\left[\frac{1}{2}, \frac{3}{4}, \frac{1}{4}\right]$	`[21,31]`
15	$\left[\frac{1}{2}, \frac{1}{4}, \frac{3}{4}\right]$	`[23,29]`
16	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[25,27]`

Table 4: Wyckoff site: `16d`, site symmetry: `-1`

No.	position	mapping
1	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[1,3]`
2	$\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{2}\right]$	`[2,28]`
3	$\left[\frac{3}{4}, \frac{3}{4}, \frac{1}{2}\right]$	`[4,26]`
4	$\left[\frac{1}{2}, \frac{1}{4}, \frac{1}{4}\right]$	`[5,15]`
5	$\left[\frac{1}{4}, \frac{1}{2}, \frac{1}{4}\right]$	`[6,24]`
6	$\left[\frac{1}{2}, \frac{3}{4}, \frac{3}{4}\right]$	`[7,13]`
7	$\left[\frac{3}{4}, \frac{1}{2}, \frac{3}{4}\right]$	`[8,22]`
8	$\left[\frac{1}{2}, 0, 0\right]$	`[9,11]`
9	$\left[\frac{1}{4}, \frac{3}{4}, 0\right]$	`[10,20]`
10	$\left[\frac{3}{4}, \frac{1}{4}, 0\right]$	`[12,18]`
11	$\left[\frac{1}{4}, 0, \frac{3}{4}\right]$	`[14,32]`
12	$\left[\frac{3}{4}, 0, \frac{1}{4}\right]$	`[16,30]`
13	$\left[0, \frac{1}{2}, 0\right]$	`[17,19]`
14	$\left[0, \frac{1}{4}, \frac{3}{4}\right]$	`[21,31]`
15	$\left[0, \frac{3}{4}, \frac{1}{4}\right]$	`[23,29]`
16	$\left[0, 0, \frac{1}{2}\right]$	`[25,27]`

Table 5: Wyckoff site: `16e`, site symmetry: `2'..`

No.	position	mapping
1	$\left[x, \frac{1}{8}, \frac{1}{8}\right]$	`[1,13]`
2	$\left[\frac{3}{4} - x, \frac{5}{8}, \frac{1}{8}\right]$	`[2,14]`
3	$\left[-x, \frac{7}{8}, \frac{7}{8}\right]$	`[3,15]`
4	$\left[x + \frac{1}{4}, \frac{3}{8}, \frac{7}{8}\right]$	`[4,16]`
5	$\left[x, \frac{5}{8}, \frac{5}{8}\right]$	`[5,9]`
6	$\left[\frac{3}{4} - x, \frac{1}{8}, \frac{5}{8}\right]$	`[6,10]`
7	$\left[-x, \frac{3}{8}, \frac{3}{8}\right]$	`[7,11]`
8	$\left[x + \frac{1}{4}, \frac{7}{8}, \frac{3}{8}\right]$	`[8,12]`
9	$\left[x + \frac{1}{2}, \frac{1}{8}, \frac{5}{8}\right]$	`[17,29]`
10	$\left[\frac{1}{4} - x, \frac{5}{8}, \frac{5}{8}\right]$	`[18,30]`
11	$\left[\frac{1}{2} - x, \frac{7}{8}, \frac{3}{8}\right]$	`[19,31]`
12	$\left[x + \frac{3}{4}, \frac{3}{8}, \frac{3}{8}\right]$	`[20,32]`
13	$\left[x + \frac{1}{2}, \frac{5}{8}, \frac{1}{8}\right]$	`[21,25]`
14	$\left[\frac{1}{4} - x, \frac{1}{8}, \frac{1}{8}\right]$	`[22,26]`
15	$\left[\frac{1}{2} - x, \frac{3}{8}, \frac{7}{8}\right]$	`[23,27]`
16	$\left[x + \frac{3}{4}, \frac{7}{8}, \frac{7}{8}\right]$	`[24,28]`

Table 6: Wyckoff site: `16f`, site symmetry: `.2'.`

No.	position	mapping
1	$\left[\frac{1}{8}, y, \frac{1}{8}\right]$	`[1,22]`
2	$\left[\frac{5}{8}, \frac{3}{4} - y, \frac{1}{8}\right]$	`[2,21]`
3	$\left[\frac{7}{8}, -y, \frac{7}{8}\right]$	`[3,24]`
4	$\left[\frac{3}{8}, y + \frac{1}{4}, \frac{7}{8}\right]$	`[4,23]`

continued ...

Table 6

No.	position	mapping
5	$\left[\frac{1}{8}, \frac{3}{4}-y, \frac{5}{8}\right]$	[5,18]
6	$\left[\frac{5}{8}, y, \frac{5}{8}\right]$	[6,17]
7	$\left[\frac{7}{8}, y+\frac{1}{4}, \frac{3}{8}\right]$	[7,20]
8	$\left[\frac{3}{8}, -y, \frac{3}{8}\right]$	[8,19]
9	$\left[\frac{1}{8}, y+\frac{1}{2}, \frac{5}{8}\right]$	[9,30]
10	$\left[\frac{5}{8}, \frac{1}{4}-y, \frac{5}{8}\right]$	[10,29]
11	$\left[\frac{7}{8}, \frac{1}{2}-y, \frac{3}{8}\right]$	[11,32]
12	$\left[\frac{3}{8}, y+\frac{3}{4}, \frac{3}{8}\right]$	[12,31]
13	$\left[\frac{1}{8}, \frac{1}{4}-y, \frac{1}{8}\right]$	[13,26]
14	$\left[\frac{5}{8}, y+\frac{1}{2}, \frac{1}{8}\right]$	[14,25]
15	$\left[\frac{7}{8}, y+\frac{3}{4}, \frac{7}{8}\right]$	[15,28]
16	$\left[\frac{3}{8}, \frac{1}{2}-y, \frac{7}{8}\right]$	[16,27]

Table 7: Wyckoff site: 16g, site symmetry: ..2

No.	position	mapping
1	$\left[\frac{1}{8}, \frac{1}{8}, z\right]$	[1,26]
2	$\left[\frac{5}{8}, \frac{5}{8}, z\right]$	[2,25]
3	$\left[\frac{7}{8}, \frac{7}{8}, -z\right]$	[3,28]
4	$\left[\frac{3}{8}, \frac{3}{8}, -z\right]$	[4,27]
5	$\left[\frac{1}{8}, \frac{5}{8}, \frac{3}{4}-z\right]$	[5,30]
6	$\left[\frac{5}{8}, \frac{1}{8}, \frac{3}{4}-z\right]$	[6,29]
7	$\left[\frac{7}{8}, \frac{3}{8}, z+\frac{1}{4}\right]$	[7,32]
8	$\left[\frac{3}{8}, \frac{7}{8}, z+\frac{1}{4}\right]$	[8,31]
9	$\left[\frac{1}{8}, \frac{5}{8}, z+\frac{1}{2}\right]$	[9,18]
10	$\left[\frac{5}{8}, \frac{1}{8}, z+\frac{1}{2}\right]$	[10,17]
11	$\left[\frac{7}{8}, \frac{3}{8}, \frac{1}{2}-z\right]$	[11,20]
12	$\left[\frac{3}{8}, \frac{7}{8}, \frac{1}{2}-z\right]$	[12,19]
13	$\left[\frac{1}{8}, \frac{1}{8}, \frac{1}{4}-z\right]$	[13,22]
14	$\left[\frac{5}{8}, \frac{5}{8}, \frac{1}{4}-z\right]$	[14,21]
15	$\left[\frac{7}{8}, \frac{7}{8}, z+\frac{3}{4}\right]$	[15,24]
16	$\left[\frac{3}{8}, \frac{3}{8}, z+\frac{3}{4}\right]$	[16,23]

Table 8: Wyckoff site: 32h, site symmetry: 1

No.	position	mapping
1	$[x, y, z]$	[1]
2	$\left[\frac{3}{4}-x, \frac{3}{4}-y, z\right]$	[2]
3	$[-x, -y, -z]$	[3]
4	$\left[x+\frac{1}{4}, y+\frac{1}{4}, -z\right]$	[4]
5	$\left[x, \frac{3}{4}-y, \frac{3}{4}-z\right]$	[5]
6	$\left[\frac{3}{4}-x, y, \frac{3}{4}-z\right]$	[6]
7	$\left[-x, y+\frac{1}{4}, z+\frac{1}{4}\right]$	[7]
8	$\left[x+\frac{1}{4}, -y, z+\frac{1}{4}\right]$	[8]

continued ...

Table 8

No.	position	mapping
9	$\left[x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[9]
10	$\left[\frac{3}{4}-x,\, \frac{1}{4}-y,\, z+\frac{1}{2}\right]$	[10]
11	$\left[-x,\, \frac{1}{2}-y,\, \frac{1}{2}-z\right]$	[11]
12	$\left[x+\frac{1}{4},\, y+\frac{3}{4},\, \frac{1}{2}-z\right]$	[12]
13	$\left[x,\, \frac{1}{4}-y,\, \frac{1}{4}-z\right]$	[13]
14	$\left[\frac{3}{4}-x,\, y+\frac{1}{2},\, \frac{1}{4}-z\right]$	[14]
15	$\left[-x,\, y+\frac{3}{4},\, z+\frac{3}{4}\right]$	[15]
16	$\left[x+\frac{1}{4},\, \frac{1}{2}-y,\, z+\frac{3}{4}\right]$	[16]
17	$\left[x+\frac{1}{2},\, y,\, z+\frac{1}{2}\right]$	[17]
18	$\left[\frac{1}{4}-x,\, \frac{3}{4}-y,\, z+\frac{1}{2}\right]$	[18]
19	$\left[\frac{1}{2}-x,\, -y,\, \frac{1}{2}-z\right]$	[19]
20	$\left[x+\frac{3}{4},\, y+\frac{1}{4},\, \frac{1}{2}-z\right]$	[20]
21	$\left[x+\frac{1}{2},\, \frac{3}{4}-y,\, \frac{1}{4}-z\right]$	[21]
22	$\left[\frac{1}{4}-x,\, y,\, \frac{1}{4}-z\right]$	[22]
23	$\left[\frac{1}{2}-x,\, y+\frac{1}{4},\, z+\frac{3}{4}\right]$	[23]
24	$\left[x+\frac{3}{4},\, -y,\, z+\frac{3}{4}\right]$	[24]
25	$\left[x+\frac{1}{2},\, y+\frac{1}{2},\, z\right]$	[25]
26	$\left[\frac{1}{4}-x,\, \frac{1}{4}-y,\, z\right]$	[26]
27	$\left[\frac{1}{2}-x,\, \frac{1}{2}-y,\, -z\right]$	[27]
28	$\left[x+\frac{3}{4},\, y+\frac{3}{4},\, -z\right]$	[28]
29	$\left[x+\frac{1}{2},\, \frac{1}{4}-y,\, \frac{3}{4}-z\right]$	[29]
30	$\left[\frac{1}{4}-x,\, y+\frac{1}{2},\, \frac{3}{4}-z\right]$	[30]
31	$\left[\frac{1}{2}-x,\, y+\frac{3}{4},\, z+\frac{1}{4}\right]$	[31]
32	$\left[x+\frac{3}{4},\, \frac{1}{2}-y,\, z+\frac{1}{4}\right]$	[32]